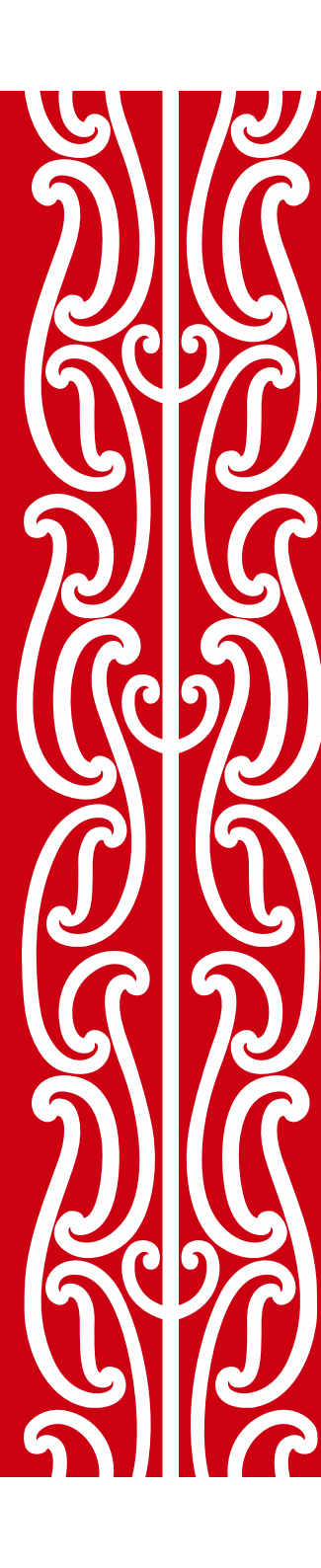




ENEL301 – Fundamentals of Engineering Economics and Management

Business Structures, Strategy, Decisions & Risk (from Week 5)

Under the Companies Act (1993), can Directors face prosecution in NZ?

- Yes. Fines, up to \$50k, depending on the offence
 - Criminal charges are possible (punishment can include fines, imprisonment)
 - Imprisonment is uncommon in NZ
 - Civil action also possible, e.g. by creditors
-
- [How we enforce the law: The action we can take against companies and directors](#)
 - [Mainzeal case](#) (Directors were found liable for trading while insolvent; Directors ordered to pay \$39.8 million)

Value creation in engineering

Design

Collaboration:

Engineering Analysis:

Sustainment:

Due diligence:

Protection:

Value creation in engineering

Design

1. Innovation, research and development
2. Product differentiation

Engineering Analysis:

3. Efficiency improvement
4. Uncertainty reduction
5. Performance forecasts

Due diligence:

6. Inspection, Checking, Testing
7. Project and Design Review
8. Compliance with standards

Collaboration:

9. Reliable coordination of other people (who then create value)
10. Teaching important skills & knowledge
11. Co-creating value with communities

Sustainment:

12. Operations, asset management and maintenance

Protection:

13. Environmental Protection
14. Security & Defence

What makes a *good* business decision?

Decision-making is always made under constraints (e.g. rules, money, time, risk)... but these help:

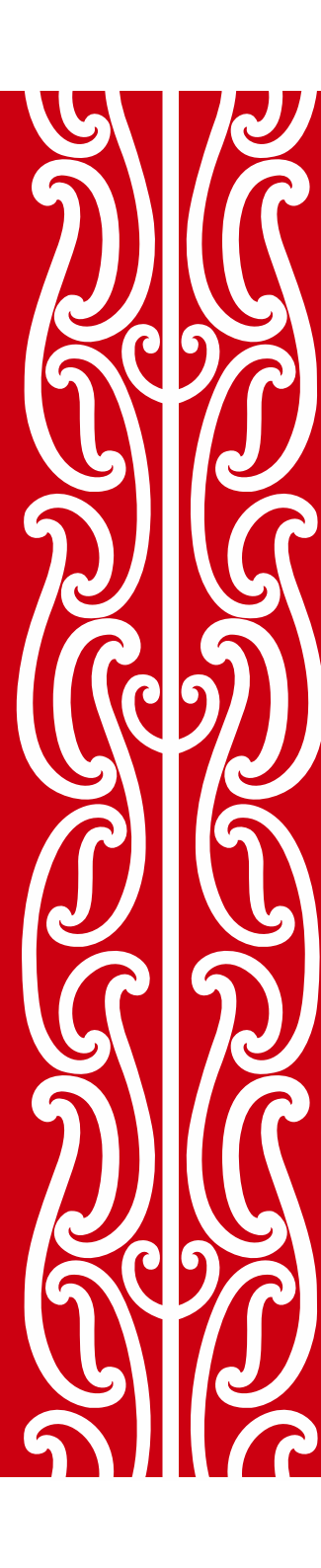
- An established process (e.g. consensus, voting, one decider)
- Trade-off between lots of factors; cost, value, opportunity, risk, ethics, stakeholder perspectives, social benefit / impact, environmental benefit / impact etc.
- Stakeholders are involved in the process
- Be transparent as possible; no surprises
- Can be 'gated' → progressive decisions based on increasingly more detailed information
 - Feasibility studies
- Bias avoidance

Coverage in ENEL301

Take-aways



- Add value; Think about each decision you make: how is what I am doing adding value?
- Prioritise your work in ways that are more likely to align with commercial objectives and build value for the communities you serve



ENEL301 – Fundamentals of Engineering Economics and Management

Sustainability

**Sustainable
development.**

Systems thinking

What we are going to cover and when...

Week	Lectures Tuesday 10-11 & Thursday 9-10	Tutorial / Workshop Weeks 1, 4-6 Wednesday 3-4 or 4-5. Weeks 2&3 Wednesday 3-5
1	Introduction to Engineering Practice & Ethics	Ethics
	Collaboration & Teamwork	
2	Māori Perspectives in Engineering	Bicultural Competence and Confidence (timetable and attend one of B01-07 OR B08-14)
	Bias, Diversity & Ethics	
3	Intellectual Property (1/2)	
	Intellectual Property (2/2)	
4	Law in New Zealand (1/2)	IP Protection
	Law in New Zealand (2/2)	
5	Data Privacy and Data Ethics	Law, Data Privacy, Data Ethics
	Business Structures, Strategy, Decisions & Risk	
6	Sustainable Development & Systems Thinking	Sustainable Development & Systems Thinking
	Life Cycle Assessment (1/2)	

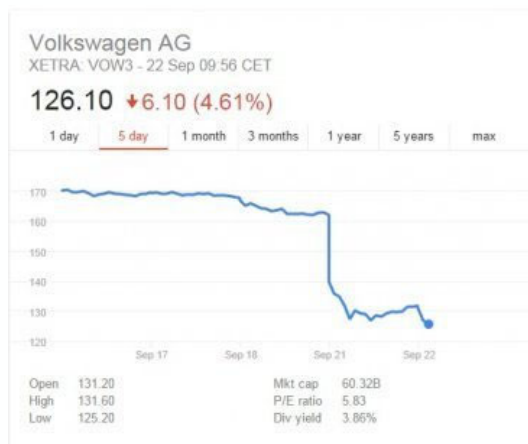
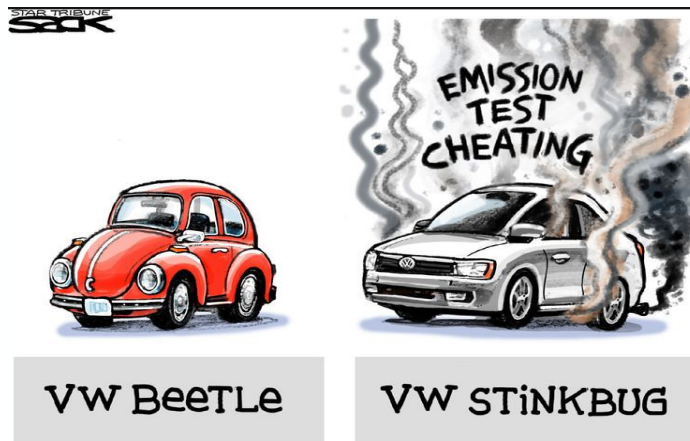
Why is this important?

Value for your company

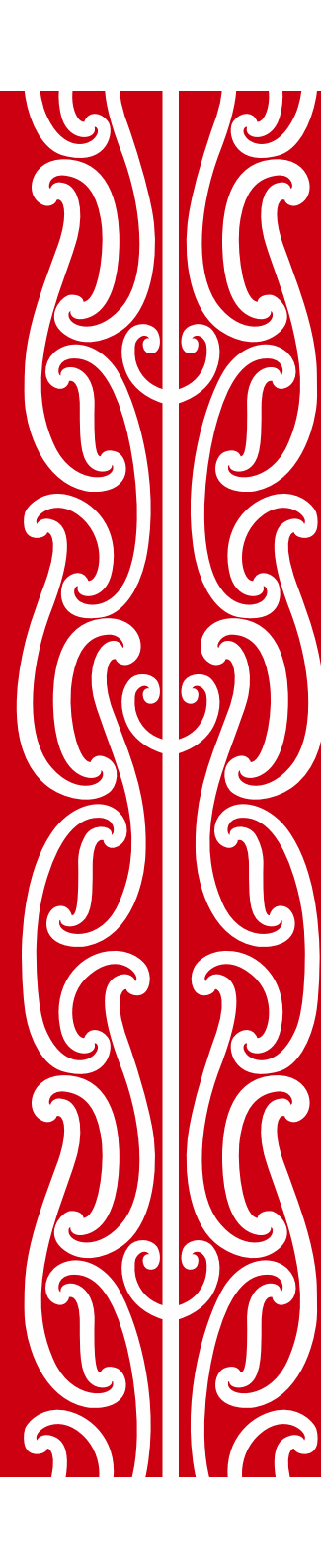
- Recognise and better manage corporate risk
 - Better decisions. Organisations do not make decisions, people do → that person will be you
 - Systems thinking can help identify outcomes from a decision
 - Minimise environmental disturbance and improve environmental outcomes
 - Better engagement with stakeholders
- Attract and retain staff when companies 'do the right thing'

Why is this important?

Value destruction



- Engineers programmed devices to cheat emissions tests
- Emissions were ~ 40 higher in real driving
- Misled regulators, customers, shareholders
- Violated environmental law
- Massive fines, recalls, executive prosecutions
- Share price collapsed

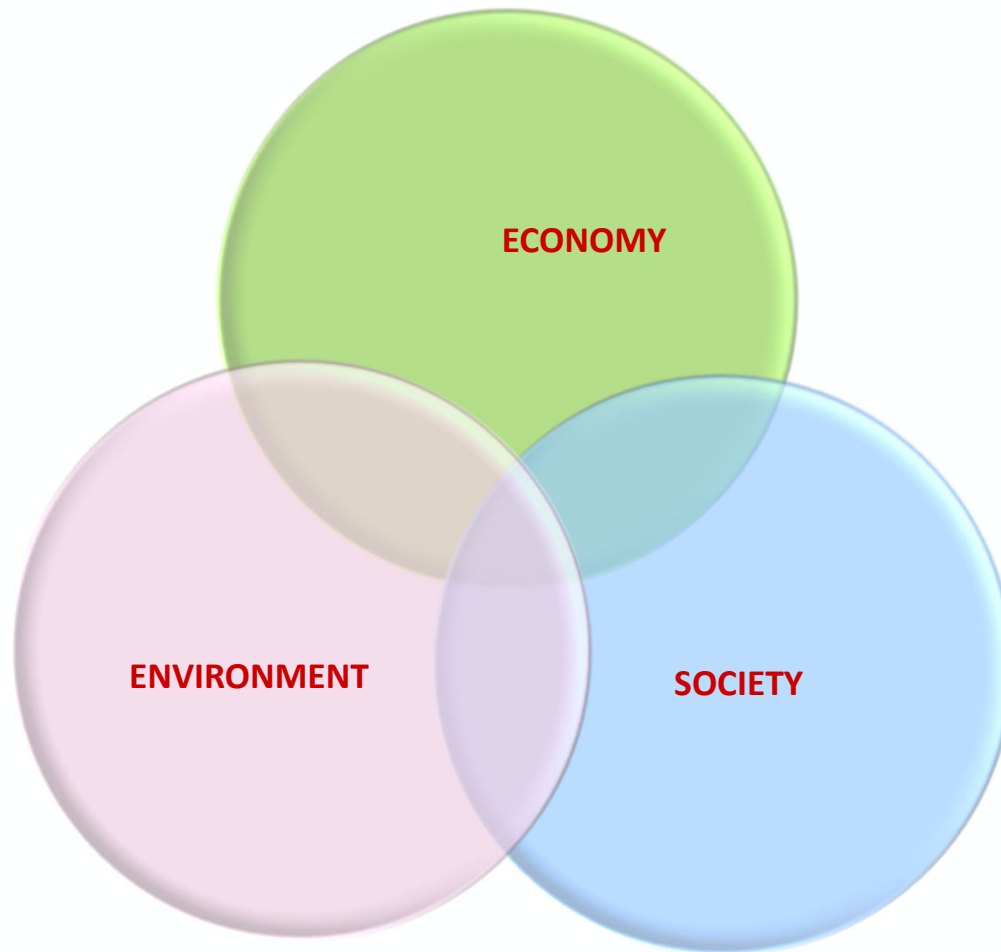


ENEL301 – Fundamentals of Engineering Economics and Management

Sustainability

Sustainable development.

What is sustainability?



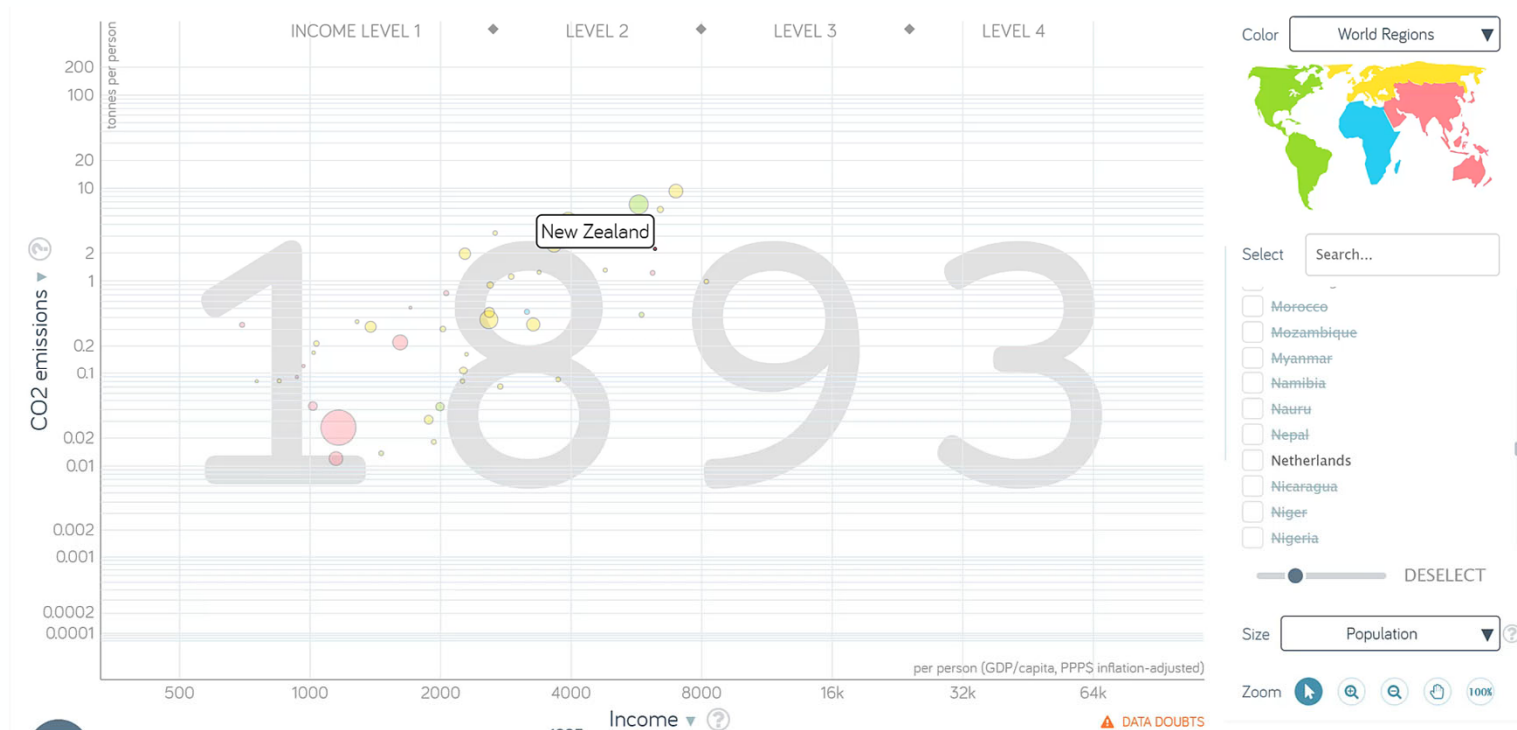
**“..meeting the needs of the present
without compromising the ability of
future generations to meet their own
needs.”**

**Brundtland Report – Our Common
Future**



GAD [CC BY-SA 3.0
(<https://creativecommons.org/licenses/by-sa/3.0>)]

Sustainable development



<https://www.gapminder.org/tools/>

Sustainable Development Goals



- A universal approach to sustainable development
- No distinction between developed and developing
- <https://sdgs.un.org/goals>
- Useful for framing, identifying (and measuring) social, environmental and economic outcomes of decisions

→ United Nations Sustainability Development Goals (SDGs) are one way of expressing sustainable development

Sustainable Development Goals Goals, Targets, Indicators

Goal

1. End poverty in all its forms everywhere



Target

1.1. By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 a day

....

Indicator

1.1.1 Proportion of the population living below the international poverty line by sex, age, employment status and geographical location (urban/rural)

....

Sustainable Development Goals

Goals, Targets, Indicators

Goal

7. Ensure access to affordable, reliable, sustainable and modern energy for all



Target

7.1 By 2030, ensure universal access to affordable, reliable and modern energy services

7.2 ...

Indicator

7.1.1 Proportion of population with access to electricity

7.1.2 Proportion of population with primary reliance on clean fuels and technology

7.2.1 ...

Sustainable Development Goals

Goals, Targets, Indicators

Goals: 17

Targets: 169

Indicators: 251

United Nations » Department of Economic and Social Affairs » Statistics Division



HOME NEWS HLG-PCCB IAEG-SDGs EVENTS **SDG INDICATORS** ▾ REPORTS ▾ RESOURCES ▾

SDG Indicators

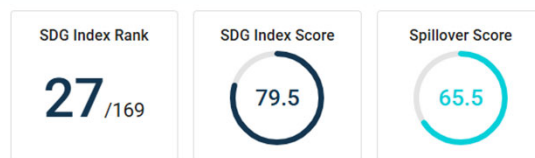
Global indicator framework for the Sustainable Development Goals and targets of the 2030 Agenda for Sustainable Development

The global indicator framework for Sustainable Development Goals was developed by the Inter-Agency and Expert Group on SDG Indicators (IAEG-SDGs) and agreed upon at the 48th session of the United Nations Statistical Commission held in March 2017.

The global indicator framework was later adopted by the General Assembly on 6 July 2017 and is contained in the **Resolution adopted by the General Assembly on Work of the Statistical Commission pertaining to the 2030 Agenda for Sustainable Development (A/RES/71/313), Annex**. According to the Resolution, the indicator framework will be refined annually and reviewed comprehensively by the Statistical Commission at its fifty-first session in March 2020 and its fifty-sixth session, to be held in 2025. The global indicator framework will be complemented by indicators at the regional and national levels, which will be developed by Member States.

<https://unstats.un.org/sdgs/indicators/indicators-list/>

Sustainable Development Goals New Zealand Context (2026 update)



SDG Dashboards and Trends

Click on a goal to view more information.



Dashboards: ● SDG achieved ● Challenges remain ● Significant challenges remain ● Major challenges remain ● Information unavailable

Trends: ↑ On track or maintaining SDG achievement ↗ Moderately improving → Stagnating ↓ Decreasing ** Trend information unavailable

Rating

- SDG achieved
- Challenges remain
- Significant challenges remain
- Major challenges remain
- [information unavailable]

Trend

- ↑ On track to achieve goal by 2030
- ↗ Score moderately increasing, insufficient to attain goal
- Score stagnating or increasing at less than 50% of required rate
- ↓ Score decreasing
- ** Trend information unavailable

<https://dashboards.sdgindex.org/profiles/new-zealand>

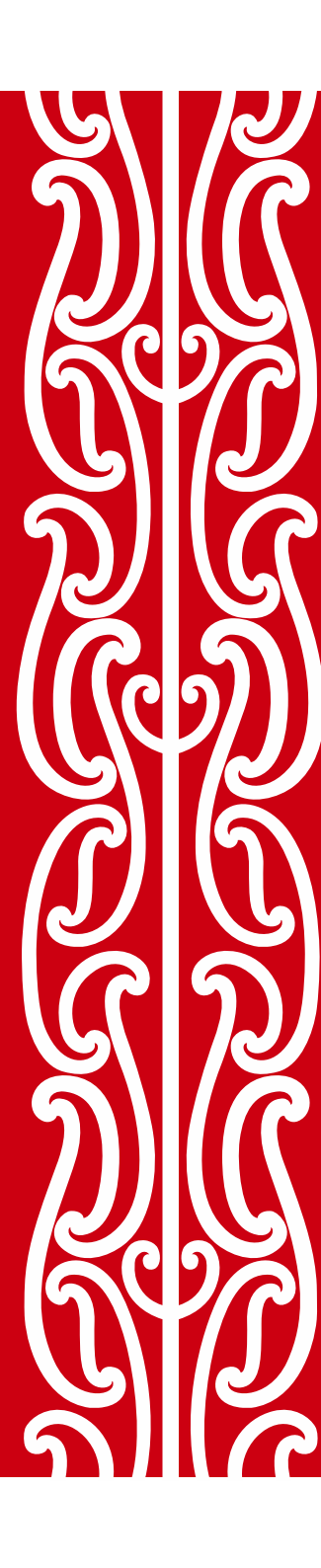
Sustainable Development Goals New Zealand Context (2026 update)

- Te Tiriti o Waitangi | The Treaty of Waitangi
- Zero Carbon Act (2019)
 - Net zero by 2050, exc. biogenic methane (agricultural, waste degradation)
 - Reducing biogenic methane emissions by 24 to 47%
 - Climate Change Commission
- Resource Management Act (1991) (under reform)
- Most codes of conduct include sustainable development type clauses (e.g. Eng NZ's 2, 3, 6, 8)
- Mandatory sustainability reporting for some engineering companies (depends on size)



Summary and take-aways

- Sustainability
 - “..meeting the needs of the present without compromising the ability of future generations to meet their own needs.”
- Sustainable development
 - Pathways to achieve economic, social and environmental outcomes
- Sustainability in New Zealand
 - Not without challenges
 - Reflected in Te Tiriti o Waitangi | The Treaty of Waitangi & law



ENEL301 – Fundamentals of Engineering Economics and Management

Sustainability

Systems thinking

Systems

-
- An assembly of components connected together in an organised way
 - The components are affected by being in the system and the behaviour of the system is changed if they leave it.
 - This organised assembly of components does something.
 - This assembly as a whole has been identified by someone who is interested in it.

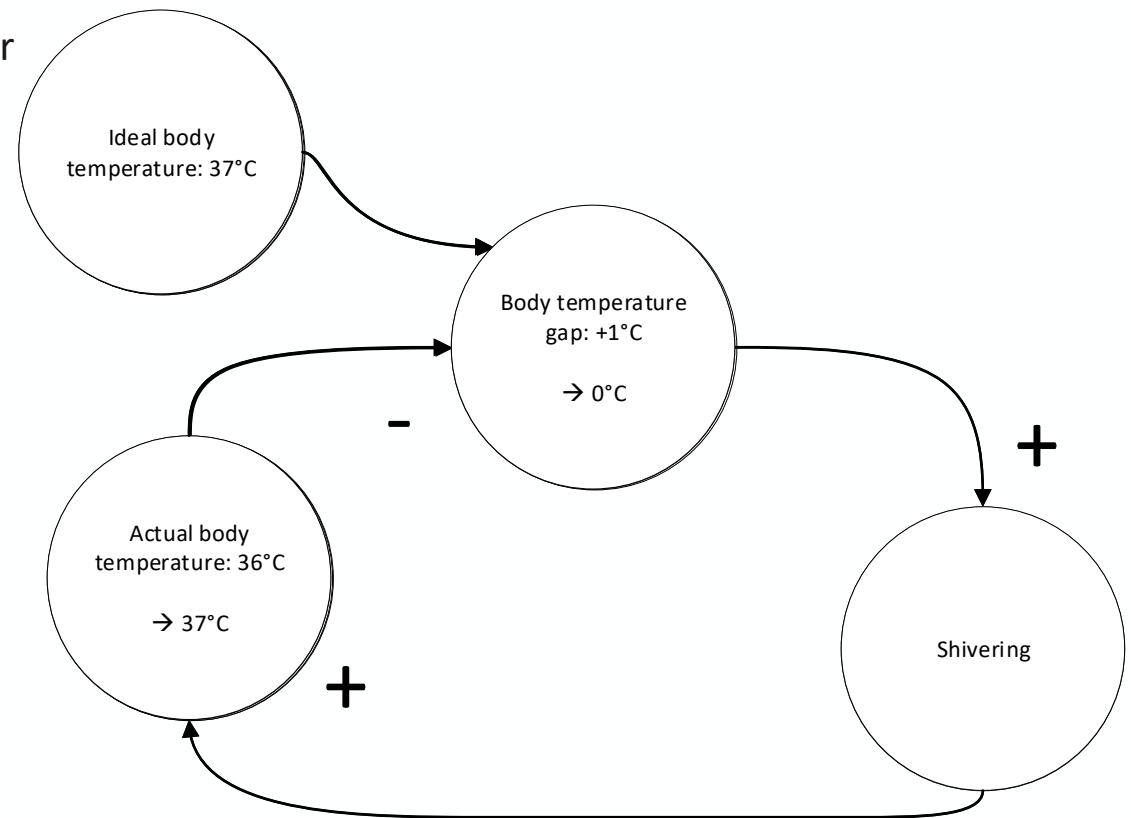
Systems thinking

- Tools to analyse, understand, model, and communicate complex systems
- Can be used to design new systems, improve old systems, diagnose problems or identify failures in existing systems
- Powerful way to address issues that are
 - Complex
 - Important
 - Chronic (not a one-time event)
 - Familiar (has a known history)
 - Persistent (people have unsuccessfully tried to solve the problem)
- A few tools that can be used to help *Be Less Wrong....*

Systems thinking

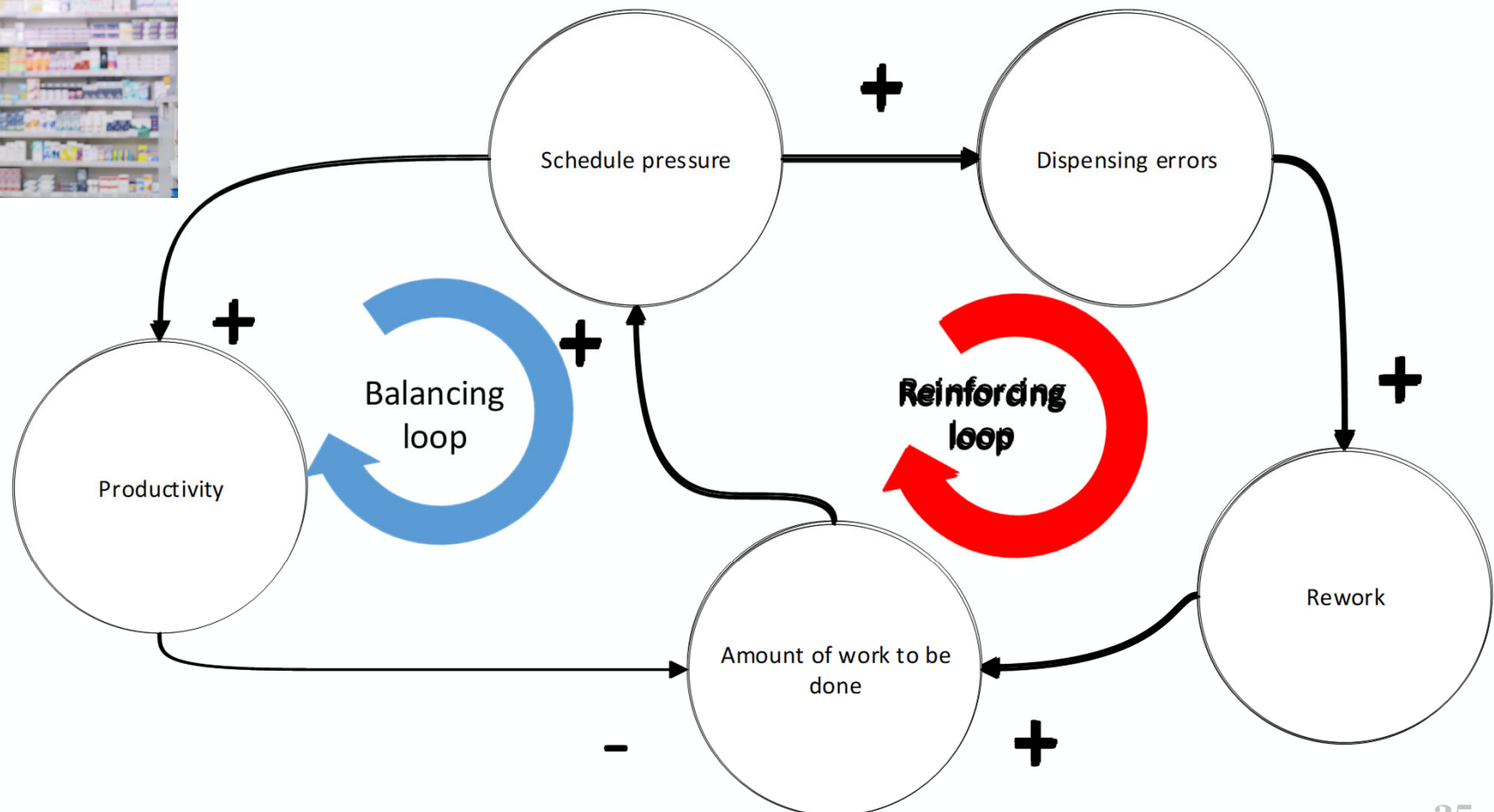
Causal loop diagrams

- Homeostasis: Tendency to resist change in order to maintain a stable internal environment
- Examples:
 - Body temperature
 - Blood sugar levels



Systems thinking

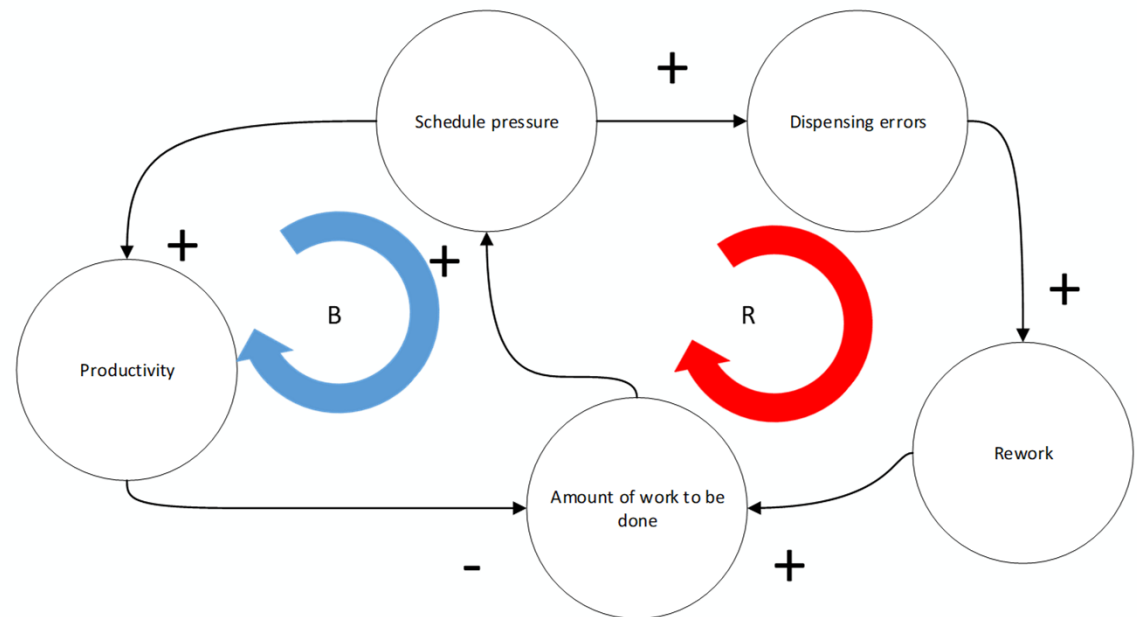
Causal loop diagrams



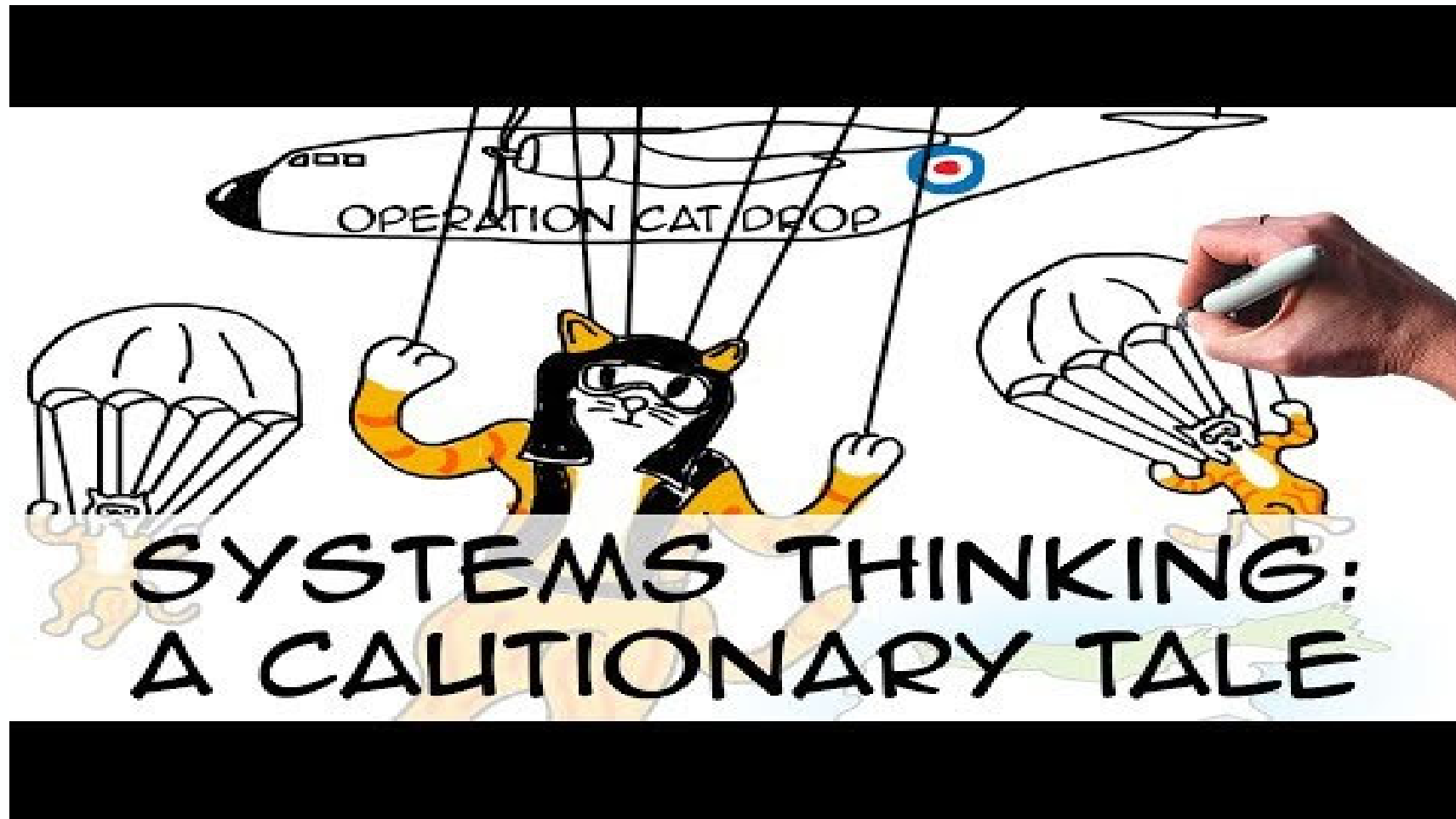
Systems thinking

Causal loop diagrams

- Create variable names → nouns
- Draw links
- Identify causal effects
 - S = Same OR +
 - O = Opposite OR -
- Identify and label any loops
 - B = Balancing
 - R = Reinforcing
- Limitation: qualitative; does not account for strength of causal effects
- Loopy: <https://ncase.me/loopy/v1.1/>

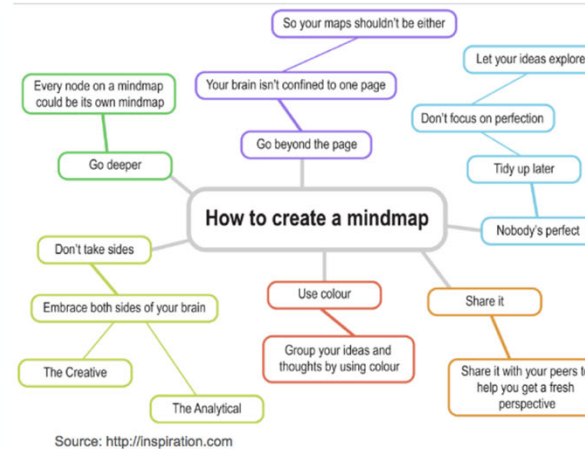


Systems thinking Operation Cat Drop



Systems thinking

Mapping tools – spray diagrams / mind maps

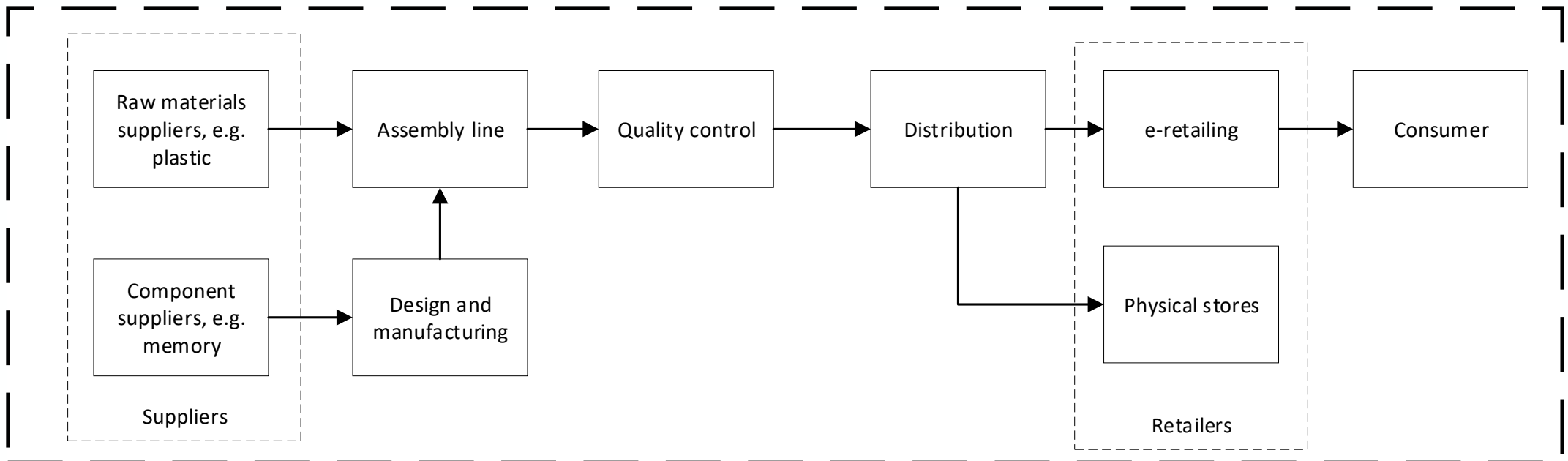


- Map of thoughts
- Useful reference for future use
- Circle central idea, potentially key nodes
- Link if there is an association

Systems thinking

Mapping tools – system map

System: Smart phone



Sustainability Assignment



Classification: In-Confidence

Sustainability Assignment

- About you: You are a part of a team of consultants to the fictitious New Zealand Data and AI Commission (NZDAIC)
- Project: Assess the proposed Datagrid hyperscale AI data centre in Makarewa, Southland.
- Evaluate whether or not the project delivers overall benefits to New Zealand, considering environmental, social and economic factors

How you assess the social and economic impacts will depend on your assumptions about who uses Datagrid's services and where those services are being applied.



Sustainability Assignment



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- 20% of ENEL301
- Teams of 4 (sometimes 5); check your ENEL301 profile in Learn
- Report due Friday 09 October at 17:00
 - One PDF, one XLSX
 1. Environmental assessment, using Life Cycle Assessment
 2. Economic Impact Assessment
 3. Social Impact Assessment
 4. Causal Loop Diagram, links to UN SD framework
- Two points of compulsory peer assessment:
 - Friday 18 September 17:00 ← preliminary feedback
 - Friday 09 October 17:00 ← used for grade adjustment

Reminders and what's next

- Workshop: Sustainable Development and Systems Thinking
 - Meet your team for the first time
 - Sustainable Development and Systems Thinking Exercises
- Friday's lecture: Sustainability; Life Cycle Assessment (1/2)